Completing the state quadratic form at a singular contact

A positive decomposition must retain the kernel at an exact zero. For the contact considered here, one event square is strictly positive at that zero. The construction below combines it with both independent state normals. Completing one normal combines its two positive coefficients. The result, together with the coefficient of the scalar form Ψ defined below, gives a positive reserve even at $n = 0$. Uniform bounds on the remaining matrix then give fixed testing functions.

Write the source coordinates as $c = (d, \lambda, n, m, C, A, D, \alpha, \beta) \geq 0$, put $D_u = 1 + u^2$, $D_v = 1 + v^2$, and define the canonical Bell slack $S_c = F_h/4$ by

$$\begin{aligned} D_u D_v S_c = {} & \lambda D_u (W + vX)^2 + n D_u D_v X^2 + m(vW - X + uvY - uZ)^2 \\ & + C D_v (uW - Y)^2 + A D_u D_v Y^2 + (D + d/2)(uW + uvX - Y - vZ)^2 \\ & + (\alpha + d/2) D_u (vY - Z)^2 + (\beta + d/2) D_v (X + uZ)^2 - \tfrac{d}{2} D_u D_v Z^2. \end{aligned} \tag{1}$$

All four state coordinates are independent. For complex states we add the same real quadratic form on the real and imaginary parts.

Theorem 1. *Take the native source defined in Appendix A, normalized by $(d, \lambda, \beta) = (1, 0, 0)$, on the domain*

$$0 < H_o \leq 2^{-12}, \quad 0 < z_o, y_o \leq 2^{-64} H_o, \quad 0 < r_o < 1, \quad 0 < t_o < 1 + z_o r_o, \quad n \geq 0. \tag{2}$$

There is a fixed finite family of testing functions, given in Table 2, whose source-dependent positive Gram matrices represent S_c on

$$0 \leq u \leq 2^{-12}, \qquad 0 \leq v \leq 2^{-25}, \qquad w\zeta \geq 0.$$

The representation is exact at $n = 0$. It implies $S_c \geq 0$ for every state and all $|u| \leq 2^{-12}$, $|v| \leq 2^{-25}$. Nonnegative additions to any of the eight event coordinates are admitted by the fixed enlarged encoding in Section 5.

The source parameters are independent of u, v; no lower bound on an opening and no comparison with a testing opening is assumed. At the native contact, with $B_s = H_o + y_o t_o$,

$$(u, v) = (\sqrt{z_o}, \sqrt{y_o}), \qquad (W, X, Y, Z) = \left(\frac{B_s}{\sqrt{z_o y_o}}, -r_o \sqrt{z_o}, \frac{H_o}{\sqrt{y_o}}, 1 \right), \tag{3}$$

we have $S_c = 0$, whereas $c_{\mathrm{test}}(aW - bY) = y_o t_o / \sqrt{(1 + z_o)(1 + y_o)} > 0$. Here $(a, b) = (u, 1)/\sqrt{D_u}$ and $(c_{\mathrm{test}}, e) = (v, 1)/\sqrt{D_v}$. Thus retaining $C[c_{\mathrm{test}}(aW - bY)]^2$ separately would lose this contact.

1 The two normals and their common reserve

Use the invertible change

$$t = \frac{X + uZ}{D_u}, \quad r = \frac{Y - uW}{D_u}, \quad w = \frac{W + uY}{D_u}, \quad \zeta = \frac{Z - uX}{D_u}; \quad (W, X, Y, Z) = (w - ur, t - u\zeta, r + uw, \zeta + ut). \tag{4}$$

The source change in Appendix A gives positive B, T, j, p, H satisfying

$$B < 2^{-11}, \quad j, T < 2^{-63}, \quad 0 < p < j, \quad 0 < H < 1, \qquad x = jB = z_o, \quad y = TB = y_o. \tag{5}$$

Put

$$N = t - pvw, \quad M = -r - Hv\zeta, \quad b_{\mathrm{c}} = vw - u\zeta/j, \quad q = v\zeta - Tuw, \quad R = \zeta - uv\, w/B, \quad R_u = v(u^2 - x)w, \tag{6}$$

$$\mu = m(1 - p), \quad h_0 = \mu + \tfrac{1}{2}, \quad M_e = \frac{n h_0 j^2}{n + h_0}, \quad \widehat{N} = \sqrt{n + h_0} \left(N + \frac{nj}{n + h_0} b_{\mathrm{c}} \right). \tag{7}$$

The positive scalar masses f_c, q_c, r_c, and the signed masses g_c, k_c, d_c, are defined explicitly in Appendix A.

Lemma 1. *Let $\Sigma = M_e + f_c/y$. Then*

$$h_0 N^2 + n(N + jb_{\mathrm{c}})^2 = \widehat{N}^2 + M_e b_{\mathrm{c}}^2, \qquad \Sigma > 0, \qquad p^2 \le 8\Sigma. \tag{8}$$

These formulas hold literally at $n = 0$.

Proof. The first identity is completion of a square. The source estimates proved below give, for $\rho = p/j < 1$,

$$\rho \le \tfrac{5}{2}(n + B), \qquad f_c/y \ge j^2 B/3.$$

If $n \le h_0$, then $M_e \ge nj^2/2$ and $p^2 = j^2\rho^2 \le (5/2)j^2(n+B) \le 5M_e + (15/2)f_c/y$. If $n \ge h_0$, then $M_e \ge h_0 j^2/2 \ge j^2/4$, so $p^2 \le 4M_e$. At $n = 0$, $M_e = 0$, $\widehat{N} = \sqrt{h_0}N$ and $\Sigma = f_c/y > 0$. No division by n or M_e occurs. □

To use this reserve, scale only by source quantities:

$$U = u/\sqrt{x}, \quad V = v/\sqrt{y}, \quad \mathcal{Z} = \zeta/\sqrt{jT}, \qquad Q_* = Uw - V\mathcal{Z}, \quad B_* = Vw - U\mathcal{Z}, \quad R_* = \mathcal{Z} - UVw,$$
$$\Psi = (U - V)^2 w^2 + \mathcal{Z}^2(1 - UV)^2 + 2UV(w - \mathcal{Z})^2, \qquad \mathcal{E} = Q_*^2 + R_*^2 + V^2(w - UV\mathcal{Z})^2. \tag{9}$$

Direct expansion gives $\mathcal{E} = (1 + V^2)\Psi - V^2 B_*^2$. Writing $\gamma = p^2/\Sigma \le 8$, we obtain

$$\gamma M_e v^4 b_{\mathrm{c}}^2 + \gamma f_c(yv^2 + v^4)\Psi - (pv^2 b_{\mathrm{c}})^2 = \gamma f_c y v^2 \mathcal{E},$$
$$\gamma T M_e u^2 v^2 b_{\mathrm{c}}^2 + \gamma T f_c u^2(y + v^2)\Psi - (\sqrt{T}\, puv b_{\mathrm{c}})^2 = \gamma T f_c y u^2 \mathcal{E}. \tag{10}$$

Both are polynomial identities in the testing variables; the second is valid at $v = 0$ without dividing the first by v^2. The signed scalar terms combine through

$$g_c V^2(\Psi - B_*^2) = g_c(\mathcal{E} - \Psi), \qquad -\kappa(VQ_* + R_*)^2 = -2\kappa V^2 Q_*^2 - 2\kappa R_*^2 + \kappa(VQ_* - R_*)^2, \quad \kappa = -k_c > 0. \tag{11}$$

These identities retain the contact while redistributing its positive masses.

2 An exact eleven-row completion

Set

$$a_N = (n + h_0)^{-1/2}, \quad b_N = \sqrt{\frac{n}{h_0(n + h_0)}}, \quad \tau = \sqrt{TH(1 - H)}, \quad \rho_Q = T\sqrt{x/q_c}, \quad \rho_R = \sqrt{jT/r_c},$$

and define the column

$$f = \begin{pmatrix} \widehat{N} \\ M/\tau \\ q/\rho_Q \\ \sqrt{M_e}\, b_{\mathrm{c}} \\ R/\rho_R \\ \sqrt{f_c}(U - V)w \\ \sqrt{f_c/(jT)}(1 - UV)\zeta \\ \sqrt{2f_c}(xy)^{-1/4}\sqrt{uv}(w - \mathcal{Z}) \\ R_u \\ pv^2 b_{\mathrm{c}} \\ \sqrt{T}\, puv b_{\mathrm{c}} \end{pmatrix}. \tag{12}$$

All its coefficients outside testing monomials are source-only. Let $Q(u, v) = \sum Q_{ab} u^a v^b$ be the symmetric matrix specified in Table 1. Each displayed off-diagonal value is a matrix entry, so its contribution to $f^T Q f$ is twice the displayed value. Define

$$\mathcal{R} = \frac{\mu p}{1 - p} N^2 + \left(\frac{\mu p}{1 - p} + h_0 + \alpha\right)(uN)^2 + \tfrac{1}{2}(vN)^2 + n[v(N + pb_{\mathrm{c}})]^2$$
$$+ (C + D + \tfrac{1}{2})(uM)^2 + (C + A + \alpha + \tfrac{1}{2})(vM)^2 + C(uvM)^2 + d_c(VQ_*)^2 + \tfrac{1}{2}p^2 y^2 (VB_*)^2$$
$$+ g_c[V(w - UV\mathcal{Z})]^2 + \kappa(VQ_* - R_*)^2. \tag{13}$$

All coefficients in $\mathcal{R}$ are nonnegative, including $d_c = H^2(A + C + \alpha + \frac{1}{2})y^2 jT + \kappa$. Put

$$\mathcal{W} = 4\Psi + 4U^2(Q_*^2 + R_*^2) - V^2(U^2 - 1)^2 w^2. \tag{14}$$

Table 1: All 44 upper-triangular records of Q. Unlisted entries vanish. Within a row each pair (i,j) has the indicated value. Write $k_R = K_R + \alpha/B$, with K_R, K_c as in (30).

Exponent	Entries of Q_{ab}
$(0,0)$ diagonal	$1,\ \tau^2(C+D+\frac{1}{2}+A),\ 1+g_c/q_c,\ 1,\ 1+(g_c-2\kappa)/r_c,$ $1-g_c/f_c-64x^2y/f_c$ (rows 6,7,8), 16, 8, 1
$(0,0)$ cross	$(1,9): a_N k_R,\ (4,9): -b_N k_R,\ (1,10): a_N/2,\ (4,10): -b_N/2,$ $(2,3): \tau\rho_Q A/T,\quad (2,11): (\mu/p)\sqrt{H(1-H)}$
$(1,0)$	$(1,5): a_N\alpha\rho_R,\quad (4,5): -b_N\alpha\rho_R$
$(0,1)$	$(2,5): \tau\alpha\rho_R$
$(2,0)$	$(3,3): -64xy/q_c,\ (5,5): -64xy/r_c,\quad (i,i): -\gamma Ty\ (i=6,7,8)$
$(0,2)$	$(3,3): -2\kappa/(yq_c),\quad (2,3): \tau\rho_Q H(C+A+\alpha+\frac{1}{2}),$ $(i,i): -8\gamma y\ (i=6,7,8)$
$(1,1)$	$(1,3): a_N(K_c/j)\rho_Q,\quad (3,4): -b_N(K_c/j)\rho_Q,$ $(1,2): (\alpha+\frac{1}{2})\tau a_N,\quad (2,4): -(\alpha+\frac{1}{2})\tau b_N$
$(2,1)$	$(2,5): -\tau BTCH\rho_R$
$(2,2)$	$(2,3): \tau\rho_Q HC,\quad (i,i): -\gamma T\ (i=4,6,7,8)$
$(0,4)$	$(i,i): -8\gamma\ (i=4,6,7,8)$

The explicit positive Gram representation of $\mathcal{W}$ is in Appendix B. The full completion is

$$D_v S_c = f^T Q f + \mathcal{R} + 16x^2y\mathcal{W} + 8\gamma f_c y v^2 \mathcal{E} + \gamma T f_c y u^2 \mathcal{E}. \tag{15}$$

One verifies this equality by substituting Table 1 and (8)–(11) into the complete two-normal expansion (31). Thus it retains both N and M and every original state direction.

3 Uniform bounds and fixed gates

Lemma 2. *On* (2)*, the matrix in Table 1 satisfies* $Q_{00} \succeq I/4$ *and*

(a,b)	$(1,0)$	$(0,1)$	$(2,0)$	$(0,2)$	$(1,1)$	$(2,1)$	$(2,2)$	$(0,4)$
$\|Q_{ab}\|_{\mathrm{op}} \le r_{ab}$	1	1	512	8	16	4	16	128.

(16)

The bounds hold for $n = 0$ *and unbounded* n*.*

Proof. The explicit source formulas in Appendix A give

$$\frac{1}{3} \le \frac{A}{T},\ THC,\ T(D+\tfrac{1}{2})(1-H),\ \frac{\alpha}{B^2},\ \frac{\mu}{jp},\ \frac{f_c}{Tx^2},\ \frac{q_c}{Tx},\ \frac{r_c}{TjB^2(1-p)} \le \frac{2}{3}. \tag{17}$$

Here a list between two bounds means that every member satisfies both. The polynomial estimate (29) gives $n(j-p)/j \le 1$, $\rho \le 5(n+B)/2$. The signed numerator K in (25) gives $0 < \mathsf{K} \le 2(p+B)$, whence

$$\frac{\kappa}{r_c} \le 3T(p+B), \quad \frac{\kappa}{q_c y} \le 4(p+B), \quad \frac{g_c}{f_c} \le 2y, \quad |K_c/j| \le 2, \quad |k_R| \le 1. \tag{18}$$

For the last two bounds use $K_c/j = p/(2j) + H(\alpha+\frac{1}{2}) - \frac{1}{2} - n(j-p)/j$ and

$$|k_R| \le \tfrac{1}{2} + \tfrac{4}{3}\delta^2 + \tfrac{8}{3}\delta < 1, \qquad \delta = 2^{-10}.$$

The normalizations therefore satisfy

$$\begin{aligned} a_N^2 + b_N^2 = 1/h_0 \le 2, \quad a_N + b_N \le 2, \quad \tau\rho_Q \le T, \quad B\rho_R < 2, \quad \alpha\rho_R \le 2B, \\ |(K_c/j)\rho_Q| \le 4\sqrt{T}, \quad (\alpha+\tfrac{1}{2})\tau \le \sqrt{T}/3, \quad TH(C+A+\alpha+\tfrac{1}{2}) < 1. \end{aligned} \tag{19}$$

Table 2: The ordered polynomial/radical list ψ of length 34. Divide every entry by $\sqrt{D_v}$ to obtain the base testing functions. The principal list $I_* = (1, 5, 10, 11, 13, 15, 16, 17, 18, 19, 20, 22, 23, 24, 27, 28)$ has length 16. Include $\sqrt{1 \pm p_{ab}}\,\psi_{I_*}/\sqrt{D_v}$ for each of the eight exponent pairs.

Factor	Multipliers, in order
t	$1, v, u, uv$
r	$1, v, u, uv$
w	$1, v, u, v^2, uv, u^2, v^3, uv^2, u^2v$
ζ	$1, v, u, v^2, uv, uv^2, u^2v, uv^3, u^2v^2$
$\sqrt{uv}\,w,\ \sqrt{uv}\,\zeta$	1 for each
$\sqrt{uw\zeta}$	$1, v, u, uv$
$\sqrt{vw\zeta}$	$1, u$

For example $\rho_Q \le \sqrt{3T}$, $\tau \le \sqrt{T}/2$, and $\alpha + \frac{1}{2} < 2/3$. Reading the absolute row sums of Table 1 gives

u	$4B$
v	B
u^2	$\max\{192B,\ 192/(1-p),\ 8Ty\}$
v^2	$\max\{1 + 8(p+B), 64y\}$
uv	$8\sqrt{T}$
u^2v	$2\sqrt{T}/3$
u^2v^2	$\max\{2/3, 8T\}$
v^4	64.

Each is below the corresponding fixed bound in (16).

For the constant matrix, rows $(1, 4, 9, 10)$ have diagonal $(1, 1, 16, 8)$ and cross columns $(a_N, -b_N)^T k_R$, $(a_N, -b_N)^T/2$. After subtracting $I/2$, their Schur correction has norm at most $2(2/31 + 1/30) = 91/465 < 1/2$. Rows $(2, 3, 11)$ have diagonal at least $(1/3, 1, 1)$: indeed

$$TH(1-H)(C + D + \tfrac{1}{2}) \ge (1-H)/3 + H/3 = 1/3.$$

Their two cross entries have absolute values at most $T, j/3$. Since $T + j/3 < 1/12$, this block is at least $I/4$. Row 5 has diagonal at least $1 - 6T(p+B) > 7/8$; rows 6–8 have diagonal at least $1 - 2y - 192B > 7/8$. These disjoint blocks prove $Q_{00} \succeq I/4$. □

Lemma 3. *The matrix Q has a positive decomposition using only the fixed gates $\sqrt{1 \pm p_{ab}}$, where $p_{ab} = (2^{12}u)^a(2^{25}v)^b$ and (a, b) runs through* (16)*.*

Proof. Set $\epsilon_{ab} = 2^{-12a-25b}$, $\alpha_{ab} = r_{ab}\epsilon_{ab}$, $A_{ab} = \epsilon_{ab}Q_{ab}$. Then $\alpha_{ab}I \pm A_{ab} \succeq 0$ and

$$Q = Q_{00} - \sum \alpha_{ab} I + \sum_{a,b} \left\{ \frac{\alpha_{ab}I + A_{ab}}{2}(1 + p_{ab}) + \frac{\alpha_{ab}I - A_{ab}}{2}(1 - p_{ab}) \right\}. \tag{20}$$

On the fixed cap, $0 \le p_{ab} \le 1$. Moreover

$$\sum \alpha_{ab} = 2^{-12} + 2^{-25} + 2^{-15} + 2^{-47} + 2^{-33} + 2^{-47} + 2^{-70} + 2^{-93} < 2^{-10}.$$

The constant reserve is therefore at least $255I/1024$. All matrices in (20) depend only on the source. □

4 Fixed functions and the other state signs

Every row of f lies in the source span of ψ_{I_*}: its eleven supports are respectively

$$\{1, 10, 20\}, \{5, 19\}, \{11, 19\}, \{10, 20\}, \{13, 18\}, \{10, 11\}, \{18, 22\}, \{27, 28\}, \{10, 17\}, \{15, 23\}, \{16, 24\}.$$

The roots in $\mathcal{R}$, $v\mathcal{E}^{1/2}$ and $u\mathcal{E}^{1/2}$ lie in the source span of ψ by (6)–(9). The seven polynomial roots of $\mathcal{W}$ in Appendix B also lie there. Its remaining roots are proportional to $\sqrt{vw\zeta}(1-U)$ and $\sqrt{uw\zeta}(1-UV, V-UV, U-UV)$, using indices 29–34. Thus (15) and (20) yield the fixed 34-dimensional base block and sixteen 16-dimensional gate blocks.

Lemma 4. *The positive-chart certificate implies the cap conclusion of Theorem 1, including its literal axes.*

Proof. Every source packet in (1) is invariant under

$$(u, v, W, X, Y, Z) \longmapsto (\varepsilon u, \eta v, W, \eta X, \varepsilon Y, \varepsilon\eta Z).$$

The rotated coordinates transform as $(t, r, w, \zeta) \mapsto (\eta t, \varepsilon r, w, \varepsilon\eta\zeta)$. Let K be the $d = 1$ core in the numerator of (1), and put $h = 1 + u^2 + v^2$, $\Delta = h^2 - u^2v^2 > 0$,

$$A_0 = \tfrac{1}{2}\begin{pmatrix} h & -uv \\ -uv & h \end{pmatrix}, \qquad B_0 = \tfrac{1}{2}\begin{pmatrix} -u^2v & -uv^2 \\ uv^2 & u^2v \end{pmatrix}.$$

Direct completion, also printed as SJ5 in the companion [1], gives

$$\begin{aligned} K/D_u = {} & \left[\begin{pmatrix} t \\ r \end{pmatrix} + A_0^{-1}B_0\begin{pmatrix} w \\ \zeta \end{pmatrix}\right]^T A_0 \left[\begin{pmatrix} t \\ r \end{pmatrix} + A_0^{-1}B_0\begin{pmatrix} w \\ \zeta \end{pmatrix}\right] \\ & + \frac{D_uD_v}{2\Delta}\left[u^2v^2(w^2 + \zeta^2) - 2uvh\,w\zeta\right]. \end{aligned} \tag{21}$$

Since $A_0 \succ 0$, $K < 0$ forces $uvw\zeta > 0$. The gauges then put $u, v > 0$ and $w\zeta > 0$, where (15) proves positivity. If $K \geq 0$, the core and all eight event packets are already nonnegative. At $u = 0$, $v = 0$, $w = 0$ or $\zeta = 0$, (21) has nonnegative remainder; no limiting division by an axis is needed. □

5 The affine encoding and its scope

A fixed enlargement makes all event additions transparent. Set

$$H_{11} = (X, Y, Z, vW, uvW, uW, uvX, vZ, u^2vW, u^3W, uZ)$$

and append to it $(uX, vX, v^2W, uv^2W, vY, uvY, W, uvZ, uY)$, obtaining H_{20}. Use $H_{20}/\sqrt{D_uD_v}$ as one base block and $\sqrt{1 \pm p_{ab}}H_{11}/\sqrt{D_uD_v}$ for

$$(a, b) = (0,1), (0,2), (1,0), (1,1), (1,2), (2,0), (2,1), (2,2), (3,0), (3,2), (4,0), (4,2), (5,0).$$

These extra gate blocks may all be zero for the weak native witness. They encode no asserted admission for $H_o > 2^{-12}$. The eight event rays have Gram witnesses in the base H_{20} alone:

λ	$e_{18} + e_{13},\ e_6 + e_7$	n	$e_1,\ e_{12},\ e_{13},\ e_7$
m	$e_4 - e_1 + e_{17} - e_{11}$	C	$e_6 - e_2,\ e_5 - e_{16}$
A	$e_2,\ e_{20},\ e_{16},\ e_{17}$	D	$e_6 + e_7 - e_2 - e_8$
α	$e_{16} - e_3,\ e_{17} - e_{11}$	β	$e_1 + e_{11},\ e_{13} + e_{19}.$

(22)

Each table entry specifies roots; their outer products are added to the corresponding source Gram. This proves the addition claim, including $d = 0$.

For clarity the affine equations can be specified without the nonlinear source dictionary. Let T_k be the nine polynomial coefficients of c_k on the right of (1), after (4). For each block let h_b be its undivided list. Its cleared product weight is 1 for the H_{20} block, $1 \pm p_{ab}$ for an H_{11} gate block, D_u for the weak base, and $D_u(1 \pm p_{ab})$ for a weak gate block. Equating every coefficient in

$$\sum_b \omega_b h_b^T G_b h_b = \sum_k c_k T_k, \qquad G_b \succeq 0, \quad c \geq 0 \tag{23}$$

defines a rational affine system. Off-diagonal Gram variables enter with factor two. Reduce products modulo $s_u^2 = u,\ s_v^2 = v,\ s_z^2 = w\zeta$, where the three symbols denote the nonnegative square roots. Their square classes are independent, so this coefficient comparison is exact on the positive open chart.

One may average the eight formal sign changes of s_u, s_v, s_z in (23). They fix all targets and weights and preserve PSD by diagonal conjugation. Cross-character entries vanish. The weak base splits as $26+2+4+2$ and each weak gate as $14+2$; the other blocks are unchanged. There are then 63 Gram blocks, total Gram order 596, 4021 Gram variables and 619 coefficient equations. Nine scalar source signs give order 605. These are counts of the specified system, not ranks or additional admission hypotheses. The literal rational arrays and their character reduction accompany this note.

The same fixed system admits finite limits of its feasible source coefficients. Indeed every block's feature list is independent on the strict chart. A common finite collection of chart evaluations therefore gives a positive definite evaluation Gram in each block. Summing (23) at these points bounds all PSD traces when the source coefficients converge; a subsequence of the Gram tuples then gives the limiting identity. This argument applies after the functions have been fixed.

Finally the cap contains the complement left by the two preserved regional certificates of [2]:

$$b > 1 - 2^{-26} \implies u^2 < (1 - 2^{-26})^{-2} - 1 < 2^{-24}, \qquad |c_{\mathrm{test}}| < 2^{-26} \implies v^2 < \frac{2^{-52}}{1 - 2^{-52}} < 2^{-50}.$$

Those certificates cover the remaining tests, including projective endpoints, on the same native envelope. Appending their unchanged order-20 and order-10 pencils gives total order 635, 4024 variables and 619 equations. Consequently the weak envelope, its nonnegative event additions and its finite coefficient limits are admitted on all tests. This does not assert admission of every canonical source or an exact lift of the full quantum set.

A Native coefficients and uniform source estimates

The following dictionary specifies the native source used above; it is the RSR source change of [1]. Put

$$\begin{aligned} B_s &= H_o + y_o t_o, \quad h_s = H_o/B_s, \quad c_s = 1 + z_o r_o, \quad R_s = 1 + h_s z_o, \\ B &= B_s R_s/c_s, \quad j = z_o c_s/(B_s R_s), \quad T = y_o c_s/(B_s R_s), \quad p = z_o(1 - r_o)/(B_s R_s), \quad H = t_o/c_s. \end{aligned} \tag{24}$$

Then $x = jB = z_o$, $y = TB = y_o$, $p/j = (1 - r_o)/c_s$ and (5) follows directly. Define the polynomials

$$\begin{aligned} K &= j + 1 - p, \quad E = 1 + B(HT + p - 1), \quad W_0 = (1 - H)j + 1 - p, \\ P &= j + 1 + B((j+1)^2 - p) + TB[Hj + j + 1 + B(Hj(2j + 2 - p) + p(2j + 1 - p))] \\ &\quad + HjT^2B^2(1 + Bp + BHj), \\ L &= K + T(W_0 - H) + BT[pK + Hj(1 - p)] + BHT^2(W_0 - 1), \\ \mathsf{N} &= p + B(-j^2 + 3jp - j - p^2 + p) + BT(Hj^2 + Hj - j^2 + jp - j + p) \\ &\quad + B^2T(2Hj^2p - Hj^2 - Hjp^2 + 2Hjp - 2j^2p + 3jp^2 - jp - p^3 + p^2) + B^2HT^2j(Hj - j + p)(1 + Bp), \\ F_1 &= -2BHjp + BHj + 2Bjp - 2Bp^2 + 2Bp - Hj - Hp - H + j + 1 - p^2/j + p/j, \\ F_2 &= -B^2H^2jp + B^2Hjp - B^2Hp^2 + B^2Hp - BH^2j - BH^2p + BHj - BHp^2/j + BHp/j, \\ F &= K + TF_1 + T^2F_2, \\ \mathsf{K} &= pW_0 + B(1 - p)[j(H(p + 1) - p) - (1 - p)(p - H)] \\ &\quad + TBH[(p + 1)(1 - H)j + (1 - p)(BHj + p + 1 - H)] + T^2B^2H^2j(1 - H). \end{aligned} \tag{25}$$

The six native weights and the scalar masses are

$$\begin{aligned} \mu &= \frac{jp(1+y)(1+x)(1+Hy)}{2P}, & m &= \frac{\mu}{1-p}, \\ n &= \frac{\mathsf{N}}{2(j-p)P}, & C &= \frac{L}{2HTP}, \\ A &= \frac{T(1+x)(1+Hy)(K+Hjy)}{2(1-HT)P}, & & \\ D &= \frac{(1+Hy)(K+Hjy)[1-y((1-H)j-p)]}{2T(1-H)P}, & \alpha &= \frac{B^2(K+Hjy)L}{2EP}, \\ f_c &= \frac{Tx^2F}{2P}, & q_c &= Ax - f_c, \\ r_c &= \frac{B^2jT(1-p)(1+x)(1+Hy)L}{2EP}, & & \\ g_c &= \frac{HT^2j^2B^3L}{2P}, & k_c &= -\frac{T^2B^2j(1+x)\mathsf{K}}{2EP}, \\ d_c &= H^2(A + C + \alpha + \tfrac{1}{2})y^2jT - k_c. & & \end{aligned} \tag{26}$$

The condition $n \ge 0$ is exactly $\mathsf{N} \ge 0$.

Here are explicit finite estimates sufficient for all bounds in the proof. For a polynomial after $p = js$, let its δ-envelope be the sum of the absolute coefficients weighted by $\delta^{\deg_B + \deg_T + \deg_j}$, with $\delta = 1/1024$ and $0 \le H, s \le 1$. The polynomials in (25) give

	$P-1$	$L-1$	$F-1$	F_1	F_2/B
envelope $\le$	$\frac{1155174955302256641}{59029581035870651712}$	$\frac{2149057539}{549755813888}$	$\frac{5633901886178307}{1152921504606846976}$	$\frac{1612711427}{536870912}$	$\frac{1077937155}{1073741824}$

(27)

The first three are below $1/128$, the last two below $4, 2$. In particular $1 \le P \le 129/128$, $127/128 \le L, F \le 129/128$, and $1 - \delta \le E \le 1$. Put $\ell = 127/128$, $h = 129/128$, $a_- = 1/(2h)$, $a_+ = (1+\delta^2)^2(1+2\delta)/[2(1-\delta)]$ and $d_+ = (1+\delta^2)(1+2\delta)(1+\delta^3)/2 + \delta/2$. Termwise substitution into (26) yields

ratio	lower	upper
A/T	a_-	a_+
THC	$\ell/(2h)$	$h/2$
$T(D+\frac{1}{2})(1-H)$	$(1-\delta^3)/(2h)$	d_+
α/B^2	$\ell/(2h)$	$(1+2\delta)h/[2(1-\delta)]$
$\mu/(jp)$	a_-	$(1+\delta^2)^3/2$
$f_c/(Tx^2)$	$\ell/(2h)$	$h/2$
$q_c/(Tx)$	$a_- - 2\delta^2/3$	a_+
$r_c/[TjB^2(1-p)]$	$\ell/(2h)$	$(1+\delta^2)^2 h/[2(1-\delta)]$

(28)

Every lower endpoint exceeds $1/3$ and every upper endpoint is below $2/3$. For example use $x, y \le \delta^2$, $1 \le K + Hjy \le 1 + 2\delta$ and $1 - \delta^3 \le 1 - y[(1-H)j - p] \le 1 + \delta^3$; for q_c use $q_c/(Tx) = A/T - x f_c/(Tx^2)$.

For the only estimate involving n, write

$$\mathsf{N} = p[1 + B(3j + 1 - p) + BT(j+1)] - Bj(j+1)[1 + (1-H)T] + \mathcal{R}_N,$$

$$\frac{\mathcal{R}_N}{B^2 T j^2} = H(2s-1) - s + s^2 + j(2Hs - Hs^2 - 2s + 3s^2 - s^3) + HT(H - 1 + s)(1 + Bjs), \quad s = p/j. \tag{29}$$

The last expression has absolute value at most $5 + 9\delta + \delta(1+\delta^2) < 6$. Thus $\mathsf{N} \ge p - 2Bj(j+1)$ and $\mathsf{N}/j \le 1 + B(3j+1) + BT(j+1) + 6B^2T(j+1) < 2$. These give $n(j-p)/j \le 1$ and $p/j \le (5/2)(n+B)$. Finally

$$\begin{aligned} \mathsf{K} - pW_0 + Bp(1-p)(j+1) &= B(1-p)[Hj(p+1) + H(1-p) + p^2] \\ &\quad + TBH[(p+1)(1-H)j + (1-p)(BHj + p + 1 - H)] \\ &\quad + T^2B^2H^2 j(1-H) \ge 0. \end{aligned}$$

It follows that $\mathsf{K} > 0$; its displayed polynomial bounds it above by $(1+\delta)p + (1 + 3\delta + 3\delta^2 + \delta^3 + \delta^4)B \le 2(p+B)$. Inserting these two bounds into (26) gives (18). All source denominators remain strictly positive; undefined zero-denominator native parameter values are not assigned values.

B The full parent and the retained circuit

For a direct verification of (15), set

$$\begin{aligned} K_c &= p/2 + j[H(\alpha + \tfrac{1}{2}) - \tfrac{1}{2}] + n(p-j), & K_R &= -\mu - (\alpha + \tfrac{1}{2})(1-p) + (T/j)K_c, \\ C_q &= A/T + v^2[CH(1+u^2) + AH + (\alpha + \tfrac{1}{2})H], \\ L_N &= njb_{\mathrm{c}} + npv^2 b_{\mathrm{c}} - j\alpha b_{\mathrm{c}} + \tfrac{1}{2}pv^2 b_{\mathrm{c}} + K_R R_u + (K_c/j)uvq, \\ L_M &= C_q q + \mu uv b_{\mathrm{c}} + v(\alpha - BTCHu^2)R, \\ H_N &= h_0(1+u^2) + \tfrac{1}{2}v^2 + \alpha u^2 + n(1+v^2), \\ H_M &= (1+u^2)[C(1+v^2) + D + \tfrac{1}{2}] + A(1+v^2) + (\alpha + \tfrac{1}{2})v^2. \end{aligned} \tag{30}$$

The scalar part and both full normals are

$$\begin{aligned} F_s &= q_c Q_*^2 + nj^2 y B_*^2 + r_c R_*^2 + d_c V^2 Q_*^2 + (np^2y^2 + \tfrac{1}{2}p^2y^2 - g_c)V^2 B_*^2 \\ &\quad + f_c\Psi + g_c V^2 \Psi + k_c \mathcal{Z}^2 (V^2 - 1)^2, \\ D_v S_c &= F_s + 2L_N N + 2L_M M + H_N N^2 + H_M M^2 + 2(\alpha + \tfrac{1}{2})uvNM + \frac{\mu p}{1-p}(1+u^2)N^2. \end{aligned} \tag{31}$$

Substitution of (26) and (4) in (1) proves (31) coefficient by coefficient, equivalently RFN8/RFN12 of [1]. The cancellations needed for this substitution, with $K_d = CH - (D + \frac{1}{2})(1 - H)$, include

$$\begin{aligned}
0 &= T(K_d + AH + \tfrac{1}{2}) - A,\\
0 &= jK_d + \mu + xTCH,\\
0 &= jK_d + xTCH + T[CH + AH + (\alpha + \tfrac{1}{2})H] - A - (\alpha + \tfrac{1}{2})(1 - p) + \alpha/B,\\
0 &= np - \mu + p/2 + [-\mu - (\alpha + \tfrac{1}{2})(1 - p)]x + j(\alpha - n) + yK_c.
\end{aligned}$$

These are rational identities in the printed dictionary; no contact equation on the state is imposed.

For completeness the RFN19 remainder has the following literal Gram form. Let

$$t_7 = \begin{pmatrix} \mathcal{Z} - U^2Vw \\ Vw - U^2Vw \\ Uw - U^2Vw \\ U\mathcal{Z} - U^2Vw \\ UVw - U^2Vw \\ UV\mathcal{Z} - U^2Vw \\ U^2w - U^2Vw \end{pmatrix}, \qquad t_3 = \begin{pmatrix} 1 - UV \\ V - UV \\ U - UV \end{pmatrix}.$$

Then

$$\mathcal{W} = t_7^T G_7 t_7 + 4Vw\mathcal{Z}(1 - U)^2 + Uw\mathcal{Z}\, t_3^T G_3 t_3, \tag{32}$$

where

$$G_7 = \begin{pmatrix} 4 & -2 & -3/2 & 0 & -1/2 & 0 & 1/2 \\ -2 & 3 & 0 & -3/2 & 0 & -1 & 2 \\ -3/2 & 0 & 4 & 3/2 & -2 & -3 & 0 \\ 0 & -3/2 & 3/2 & 4 & -3/2 & 0 & -3/2 \\ -1/2 & 0 & -2 & -3/2 & 3 & 2 & -1 \\ 0 & -1 & -3 & 0 & 2 & 8 & -4 \\ 1/2 & 2 & 0 & -3/2 & -1 & -4 & 4 \end{pmatrix}, \qquad G_3 = \begin{pmatrix} 3 & -2 & -2 \\ -2 & 2 & 2 \\ -2 & 2 & 3 \end{pmatrix}.$$

Their leading principal minors are respectively $(4, 8, 101/4, 1049/16, 907/16, 3279/16, 743/16)$ and $(3, 2, 2)$. Thus both matrices are positive definite. Since $U, V \geq 0$ and $w\mathcal{Z} \geq 0$, (32) is a sum of squares in precisely the radicals included in Table 2.

References

[1] *Finite semidefinite representation of the* $(2, 2, 2)$ *quantum set: proof companion*, preserved source `latex/certificates/c2_short.tex`, source dictionary RSR2–RSR19, full-parent identities RFN4–RFN12, RFN18b–RFN19, and sign completion SJ1–SJ7 (component `pf77de02d80c4`).

[2] *Two fixed regional completions for the C2 source*, `extracted-lifts/canonical-completion/fixed-c2-gates/Fixed_C2_Completions.tex`, regional identities and native-envelope estimates.